

MERGE_{NET}: DIFFERENTIABLE SPATIAL ROUTING WITH A PHYSICAL TOKEN BOTTLENECK

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ABSTRACT

Modern visual recognition requires models to capture both fine-grained image detail and long-range context. Vision transformers meet these needs by representing images as patch tokens and modeling their interactions, but retaining every token across layers becomes costly at high resolution. Token merging reduces this cost by consolidating redundant tokens. Yet its success depends both on learning which tokens to combine without losing task-relevant information and on translating those decisions into a physically shorter sequence. DTEM learns the merging policy with dedicated merge embeddings and a continuous relaxation. However, its soft training path retains every token slot: it learns merge decisions without realizing physical compression, which still requires hard merging. MergeNet closes this gap with a local-to-latent architecture. Differentiable routing consolidates information according to the learned merge decisions; fixed-budget selection then physically shortens the sequence before global reasoning. To make this transition trainable and practical, we combine differentiable budgeted selection with fused, mass-aware attention over merged tokens. We evaluate MergeNet on ImageNet-1K image classification, using DeiT-S as the dense baseline and comparing against DTEM, ToMe, and PiToMe at matched token budgets. At a $2\times$ carrier reduction, the trained MergeNet checkpoint obtains 81.360% top-1, compared with 82.246% for dense DeiT-S/8, 81.932% for post-training ToMe, and 81.742% for PiToMe; the matched DTEM result remains unavailable. Within MergeNet, radius-3 routing improves a matched 150-epoch ImageNet control by 1.670 percentage points over global routing and 0.598 points over flattened windows. A three-seed CIFAR-100 study retains this spatial advantage against global, flattened-window, and degree-matched receiver controls. These results support the spatial routing mechanism and physical bottleneck, while trained-checkpoint latency, memory, and component ablations remain necessary before making an accuracy–efficiency frontier claim.

1 INTRODUCTION

Vision transformers use patch tokens to represent image detail and self-attention to integrate information across the image. Processing a fine patch grid through every block, however, makes this global reasoning expensive. Token merging reduces the sequence by combining patches rather than simply removing them (Bolya et al., 2023). A learned merger thus faces two linked requirements: learn where information should flow, and physically reduce the sequence seen by downstream blocks during training.

Differentiable merging does not automatically solve this execution problem. DTEM learns dedicated merge embeddings through continuous grouping and merging (Lee & Hong, 2024). Its relaxed operators retain all token slots; for end-to-end training, DTEM alternates soft updates of the merge embeddings with hard, token-reducing updates of the backbone. Thus DTEM uses both continuous policy learning and physical compression during training, but in separate update paths. We ask whether one forward pass can propagate gradients through soft routing while downstream blocks process an exact, physically shorter sequence.

We study this question with MergeNet, a local-to-latent vision transformer. Six local blocks encode the full patch grid, after which soft mass transport learns how patch information moves between

slots. A hard gather then selects K carriers from N patches, so the subsequent latent blocks receive a physically shorter tensor during both training and inference. A residual cross-attention read lets those carriers access the full-grid local features before global latent reasoning. The gather indices are discrete; a straight-through weight connects the selected features to continuous routing scores without claiming to differentiate the indices themselves.

The routing support follows the original two-dimensional patch grid. A donor can send mass only to eligible receivers within radius R , while learned similarities determine the weights and the resulting mass determines which carriers survive the gather. This constraint tests whether spatial proximity is useful for learned transport, rather than assuming that a fixed pooling window is sufficient. We test this prior against global, flattened-window, and degree-matched routing controls.

Carrier mass also has a defined role after selection. Following proportional attention in ToMe (Bolya et al., 2023), MergeNet adds each carrier’s $\log m$ to its attention key logits. This per-key bias is rank one across queries; the released operator reproduces it by adding one query/key dimension and using the original softmax scale in an unmodified FlashAttention call (Dao et al., 2022). We use an existing threshold-based smooth top- k relaxation (Su, 2024) for the continuous selection scores. Our contribution is the way these components connect soft spatial transport to an exact-budget latent path, together with the mass-bias implementation.

The experiments support the spatial-routing mechanism, with a clear limit on the overall performance claim. In a single-seed, 150-epoch ImageNet-1K study, radius-3 routing improves top-1 accuracy by 1.670 percentage points over global routing and 0.598 points over a flattened-window control. A three-seed CIFAR-100 study also favors radius 3 over global, flattened, and degree-matched routing. At a matched final budget of 392 patches, the trained 300-epoch MergeNet checkpoint reaches 81.360% ImageNet top-1, below post-training ToMe’s 81.932%; the uncompressed DeiT-S/8 reaches 82.246%. The current results therefore establish the behavior of the proposed routing and physical bottleneck, but do not establish an accuracy–latency frontier; matched DTEM accuracy and trained-checkpoint timing remain to be measured.

Our contributions are:

- a local-to-latent architecture that combines differentiable spatial mass transport with an exact-budget gather before latent processing;
- controlled evidence for original-grid routing, including a degree-matched, three-seed CIFAR-100 comparison; and
- an exact query/key reparameterization for mass-aware attention with an unmodified FlashAttention call, accompanied by separate reporting of accuracy, token budget, arithmetic cost, and measured runtime.

2 RELATED WORK

Token reduction. DynamicViT learns token retention (Rao et al., 2021), while A-ViT assigns adaptive depth through halting (Yin et al., 2022). ToMe uses bipartite soft matching and can compress pretrained or jointly trained transformers (Bolya et al., 2023); PiToMe (Tran et al., 2024), MCTF (Lee et al., 2024), and SAD-TM (Xie et al., 2026) modify its matching signal or merge schedule. DiffRate treats layerwise budgets as another learned variable (Chen et al., 2023). Equal final counts therefore do not imply equal layerwise computation.

Differentiable merging. DTEM is the closest predecessor: it decouples the merge representation from backbone features and trains it through relaxed grouping (Lee & Hong, 2024). MergeNet builds on its OpenToMe lineage (Westlake-AI, 2025), but places continuous mass transport and a hard fixed-budget gather in one forward path. The soft top- k relaxation follows Su (2024); it is not claimed as a new selection operator. Proportional attention using token size is also established by ToMe. Our systems addition is the rank-one query/key reparameterization for retaining that bias in FlashAttention.

Spatial structure and recovery. DSM and ALGM combine local and global merging (Heo et al., 2024; Norouzi et al., 2024); GTP-ViT propagates tokens over spatial and semantic edges (Xu et al., 2024). Recent reducers constrain matching along rows/columns or after space-filling reorder-

ing (Gong & Lis, 2025; Li & Xu, 2025). MergeNet instead constrains eligible routes by Euclidean radius on the original grid. LookupViT uses cross-attention between compressed and high-resolution streams (Koner et al., 2024); our recovery read is an architectural component rather than a claimed general recovery method. Since local merging can reduce arithmetic without improving throughput (Heo et al., 2024), we keep analytic cost and measured runtime distinct.

3 METHOD

3.1 OVERVIEW

MergeNet connects differentiable routing to a physically shorter latent sequence within one forward pass. Given N image patches and a class token, a local encoder produces full-grid features $H^{(1)}, \dots, H^{(L)}$. Starting from $X^{(0)} = H^{(L)}$ and unit patch masses, L soft routing steps redistribute information while retaining every patch slot. A hard gather then selects K patch carriers. The carriers read the pre-transport features through cross-attention, and a latent encoder followed by a class head produces the prediction. Figure 1 shows the project schematic; the evaluated classifier uses its encoder-to-latent path.

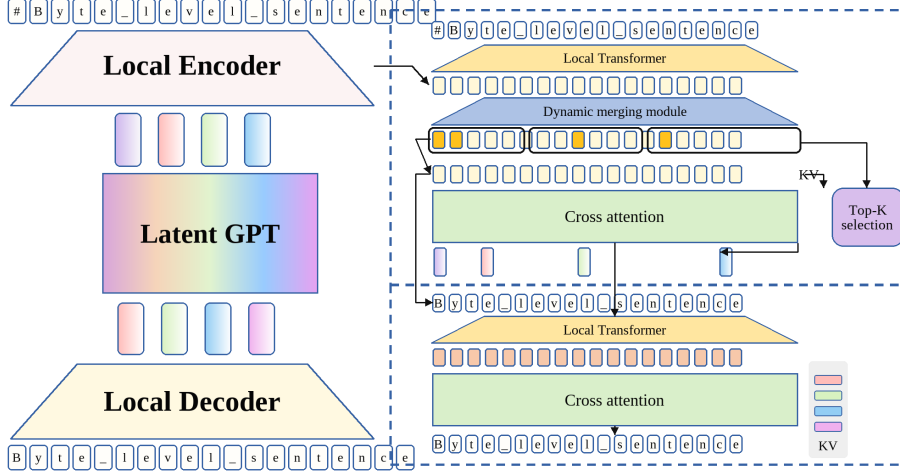


Figure 1: Project schematic of the local-to-latent design. The evaluated ImageNet classifier uses the encoder, dynamic merger, top- K selection, recovery cross-attention, latent encoder, and class head. The generic local decoder branch shown here is not used in this classifier.

3.2 SPATIAL MASS ROUTING

Spatial routing restricts where patch information may move without fixing the routing weights. Patch i keeps its coordinate u_i on the original two-dimensional grid. At routing step ℓ , patches are partitioned into disjoint donor and receiver sets A_ℓ and B_ℓ ; the class token participates in neither set. The eligible receivers for donor i are

$$\mathcal{E}_i^{(\ell)}(R) = \{j \in B_\ell : \|u_i - u_j\|_2 \leq R\}. \quad (1)$$

Coordinates remain tied to original slots as features move, so this rule constrains individual transport edges rather than the model’s complete receptive field.

The routing weights combine learned similarity with a soft donor decision. A learned projection of $H^{(\ell)}$ yields normalized metrics $z_i^{(\ell)}$. For eligible pairs, we compute

$$s_{ij}^{(\ell)} = \langle z_i^{(\ell)}, z_j^{(\ell)} \rangle / \tau, \quad q_{ij}^{(\ell)} = \frac{\exp s_{ij}^{(\ell)}}{\sum_{r \in \mathcal{E}_i^{(\ell)}(R)} \exp s_{ir}^{(\ell)}}. \quad (2)$$

The row energy $e_i^{(\ell)} = \sum_{j \in \mathcal{E}_i^{(\ell)}(R)} \exp s_{ij}^{(\ell)}$ enters a threshold-based smooth top- k operator (Su, 2024). Given the step budget k_ℓ , it produces a continuous donor weight $g_i^{(\ell)}$; the transport weight is $a_{ij}^{(\ell)} = g_i^{(\ell)} q_{ij}^{(\ell)}$ on eligible edges and zero elsewhere; a donor with no eligible receiver has zero assignment. The smooth weights provide a gradient path from the task loss to the routing metrics.

Soft transport updates mass and receiver features without removing the donor slot. With patch feature X_i and mass m_i , one step gives

$$\begin{aligned} m'_i &= m_i \left(1 - \sum_j a_{ij} \right), & X'_i &= X_i, \\ m'_j &= m_j + \sum_i a_{ij} m_i, & X'_j &= \frac{m_j X_j + \sum_i a_{ij} m_i X_i}{m'_j}. \end{aligned} \quad (3)$$

The sums cover eligible edges in the current donor–receiver partition. The implementation stabilizes residual fractions and denominators, while all patch slots remain present for the next routing step.

3.3 PHYSICAL BOTTLENECK AND RECOVERY

A hard gather turns the soft routing state into the shorter sequence used by later blocks. For compression factor λ , the patch-carrier budget is

$$K = N - \lfloor N(\lambda - 1)/\lambda \rfloor. \quad (4)$$

Let $\mathcal{I} = \text{TopK}(m_{\text{patch}}^{(L)}, K)$ contain the indices of the K largest patch masses, ordered before gathering. After normalizing the routed features, $\bar{X} = \text{Norm}(X^{(L)})$, we form the carrier sequence $C = [\bar{X}_{\text{cls}}; \bar{X}_{\mathcal{I}}]$. Thus C contains exactly $K + 1$ tokens, including the protected class token, in both training and inference.

The gather keeps its indices discrete while passing gradients to soft selection scores. A second smooth top- K operator applied to the final patch masses gives weights h_i . Each selected feature is multiplied by $1 + h_i - \text{stopgrad}(h_i)$, which equals one in the forward pass. This connects the selected features to the continuous scores without differentiating the index choice.

Recovery lets the carriers read information still available on the full grid. Using $H^{(L)}$ as the pre-transport key and value sequence, define

$$Z = C + \text{CrossAttn}(C, H^{(L)}, H^{(L)}). \quad (5)$$

The latent encoder processes Z , and its class-token output enters the classification head. Global latent self-attention therefore runs on $K + 1$ tokens, while recovery still reads the full-grid features.

3.4 MASS-AWARE LATENT ATTENTION

The selected carriers represent different amounts of transported mass. Following proportional attention in ToMe (Bolya et al., 2023), latent self-attention adds the log of each carrier mass to its key logit. For per-head query, key, and value matrices Q_a, K_a, V_a , head width d_h , and selected masses m_j , this is

$$\text{Attn}(Q_a, K_a, V_a; m) = \text{softmax} \left(\frac{Q_a K_a^\top}{\sqrt{d_h}} + \mathbf{1} \log(m)^\top \right) V_a. \quad (6)$$

The class token retains unit mass. This bias accounts for mass in attention; its contribution to classification accuracy requires a separate ablation.

The per-key bias can be encoded as one extra query/key dimension. Set $\hat{Q}_a = [Q_a, \sqrt{d_h} \mathbf{1}]$ and $\hat{K}_a = [K_a, \log m]$. With the original scale $1/\sqrt{d_h}$,

$$\frac{\hat{Q}_a \hat{K}_a^\top}{\sqrt{d_h}} = \frac{Q_a K_a^\top}{\sqrt{d_h}} + \mathbf{1} \log(m)^\top. \quad (7)$$

The released FlashAttention-compatible operator implements this identity after padding the head dimension. The reported ImageNet classifier uses the same bias through PyTorch’s scaled-dot-product attention; the operator identity alone does not establish a runtime gain.

4 EXPERIMENTAL PROTOCOL

Training and evaluation. We train on ImageNet-1K and report EMA top-1 on its 50,000-image validation split. Main 224-pixel runs use AdamW for 150 or 300 epochs with global batch 1,024, cosine decay, five warmup epochs, weight decay 0.05, label smoothing 0.1, Mixup 0.8, CutMix 1.0, RandAugment, random erasing 0.25, and no teacher. The matched geometry controls use learning rate 5×10^{-4} ; the selected MergeNet configuration uses 7.5×10^{-4} . Thus geometry comparisons are recipe matched, whereas the final dense-MergeNet comparison uses each selected configuration. Full arguments and the run inventory are in Appendix A.

The 384-pixel models fine-tune the 300-epoch EMA weights for 30 epochs; the 512-pixel models add 15 epochs. All retain effective global batch 1,024 via gradient accumulation. We report both the best and final EMA values because best-checkpoint selection uses the same validation split. The latest supplied packet contains 23 complete ImageNet schedules and one abandoned 140/150 weight-decay probe. We recompute endpoints by header name from 3,560 contiguous summary rows and cross-check 3,354 rows against supplied logs; missing-log rows remain summary-only. The company checkpoints were not transferred, so this is a record audit rather than an independent checkpoint replay.

Compression comparisons. The main table fixes 392 final patches at 224 pixels. ToMe and PiToMe are applied after training to the dense checkpoint, while MergeNet is trained with its native bottleneck. The budget sweep evaluates the complete validation set at five budgets. These rows compare deployable checkpoints; they do not isolate the routing rule under identical training. In the routing controls, global permits any receiver, flat-8 uses a width-8 window in row-major token order, and radius 3 limits receivers by Euclidean distance on the original patch grid. A separate 12-run CIFAR-100 study adds a receiver-permuted radius-3 graph that preserves each donor’s candidate count and the receiver degree multiset while disrupting two-dimensional locality. It evaluates all four geometries over three seeds for 200 epochs; its preregistered endpoint is epoch-199 EMA top-1.

Timing. Historical 224- and 384-pixel latency records use synthetic inputs and random weights; one later ToMe sweep also disables proportional attention. We report these only as architecture diagnostics. The primary table leaves latency and memory blank until dense, matched DTEM, ToMe, and MergeNet are measured from trained checkpoints with identical preprocessing, precision, batch, attention policy, token hooks, and CUDA-event timing. The same collaborator packet should supply two inference-only component scores and passive routing traces. The public repository intentionally distributes source and configuration rather than trained weights; the headline result is therefore supported by the record audit above rather than public checkpoint replay.

5 RESULTS

5.1 PRIMARY MATCHED-BUDGET COMPARISON

Table 1 fixes the final budget at 392 patches, excluding CLS. The MergeNet row is the 50,000-image re-evaluation (81.360%); its training summary reports a best EMA value of 81.368%. Dense DeiT-S/8 reaches 82.246%, while post-training ToMe and PiToMe reach 81.932% and 81.742%, respectively. Thus the current evidence establishes a learnable physical bottleneck, but not a stronger accuracy frontier. The matched DTEM score and common-protocol systems cells remain explicit blanks.

5.2 DOES THE SPATIAL PRIOR HELP?

The 150-epoch ImageNet controls isolate routing support at learning rate 5×10^{-4} . Radius 3 reaches 78.888%, improving over global routing by 1.670 points and flattened windows by 0.598 (Table 2). Radius 2 is only 0.054 points higher, so one seed does not identify an optimal radius. The global arm is an architectural control with MergeNet’s local/latent split, not a reproduction of DTEM.

Radius changes both geometry and candidate count. We therefore repeat the comparison on CIFAR-100 with a receiver permutation that preserves each realized receiver degree. Across three seeds,

Table 1: ImageNet-1K at 224 pixels and a 392-patch target. Accuracy is percent. Blank cells are requested measurements, not zero values or estimates.

Method	Compression path	Input→final	×	Top-1 (%)	Median ms	Peak GiB
DeiT-S/8	dense, trained	784 → 784	1.00	82.246	_____	_____
DTEM	matched run	784 → 392	2.00	_____	_____	_____
DeiT-S/8 + ToMe	post-training	784 → 392	2.00	81.932	_____	_____
DeiT-S/8 + PiToMe	post-training	784 → 392	2.00	81.742	_____	_____
MergeNet ($R = 3$)	end-to-end	784 → 392	2.00	81.360	_____	_____

Table 2: ImageNet routing controls. All rows use the same 150-epoch recipe; accuracies are single-seed EMA top-1.

Routing / probe	LR (10^{-4})	Best (%)	Final (%)
Global	5.0	77.218	77.216
Flat window 8	5.0	78.290	78.278
Spatial $R = 2$	5.0	78.942	78.942
Spatial $R = 3$	5.0	78.888	78.888
Spatial $R = 3$	7.5	79.316	79.316

radius 3 ranks first among the four geometries (Table 3). Every one of the nine paired radius-3 minus control differences is positive. Against the degree-matched control, the mean gain is 1.42 ± 0.30 points. These runs support a locality effect beyond degree, but do not transfer ImageNet uncertainty.

Table 4 ties each mechanism claim to a control. The two remaining component interventions require only one inference pass of the collaborator-held final checkpoint; they are diagnostic ablations rather than separately trained architectures.

5.3 LONG SCHEDULES, RESOLUTION, AND BUDGETS

At 300 epochs, MergeNet trails dense DeiT-S/8 by 0.878 points at 224 pixels (Table 5). Fine-tuning narrows this gap to 0.394 at 384 pixels; at 512 pixels the gap is 0.450, so the three points do not support a monotonic convergence claim. Different MergeNet radii are used across this ladder and the 512-pixel arm descends from the radius-scaled 384 checkpoint.

Changing only the evaluation-time carrier budget gives MergeNet top-1 values of 81.380, 81.360, 81.074, 80.850, and 80.228% at 523, 392, 314, 262, and 196 patches. ToMe applied to the dense checkpoint is higher at every one of these 224-pixel budgets, with gaps from 0.572 to 1.524 points; the same ordering holds at all five 384-pixel budgets. This is a clear negative result for the current end-to-end model. It is not a controlled comparison of training objectives, because ToMe starts from the stronger dense checkpoint. Detailed curves, optimizer probes, and complete schedules are in the appendix.

6 EXECUTION EVIDENCE

A code-derived accounting estimates 14.13 GMAC per image for MergeNet and 22.41 GMAC for dense DeiT-S/8 at 224 pixels, a 36.9% reduction. This includes projections, MLPs, attention, routing transport, and recovery, but omits sorting, indexing, and data movement. Since all six local blocks remain dense and recovery costs $O(NK)$, a $2\times$ final carrier reduction does not imply a $2\times$ model reduction.

Existing runtime records confirm this distinction. On one H20-3e with random weights, fp16, synthetic batch 64, and a guarded CUDA-event harness, MergeNet takes 45.71 ms versus 42.68 ms for dense DeiT-S/8 at 224 pixels and uses more peak allocated memory. At 384 pixels it takes 136.54 versus 170.61 ms, a 20.0% reduction. A local A100 reproduction does not recover this crossover. These architecture benchmarks are useful boundary checks, but they cannot be paired with trained-checkpoint accuracy. The primary table therefore reserves common-protocol timing rather than reporting an unsupported Pareto frontier.

Table 3: CIFAR-100 epoch-199 EMA top-1 (%), mean $\pm$ sample standard deviation over seeds 42, 43, and 44.

Endpoint	Global	Flat 8	Degree-matched	$R = 3$
EMA top-1	68.48 $\pm$ 0.10	69.29 $\pm$ 0.54	69.17 $\pm$ 0.62	70.59 $\pm$ 0.88

Table 4: Claim-to-ablation map. “Done” rows contain complete experiments.

Claim	Control	Result
2-D routing	global, flat, $R = 2/3$	Done; Table 2
Locality beyond degree	receiver permutation	+1.42 $\pm$ 0.30 points
Budget behavior	20 matched token settings	ToMe higher in 20/20
Mass-aware attention	set log m bias to zero	[pending: score]
Full-grid recovery	bypass cross-attention	[pending: score]

7 DISCUSSION AND LIMITATIONS

The experiments separate a positive mechanism result from a negative system result. Original-grid support improves over global, flattened, and degree-matched controls, so the spatial prior is useful within this architecture. The complete model nevertheless remains below dense DeiT-S/8 and ToMe in recorded top-1, and it is slower at 224 pixels in the available random-weight benchmark. The current evidence therefore supports “trainable spatial routing with a physical bottleneck,” not a state-of-the-art accuracy–efficiency frontier.

The design also retains substantial full-grid work. Soft transport does not shorten the local stage; recovery reads all local features; and sparse scoring still incurs indexing and initialization costs. A local edge constraint does not bound the final receptive field after multiple transports, nor does it establish semantic boundary preservation. The 150-epoch ImageNet geometry study has one seed, while the replicated degree-matched evidence is on CIFAR-100. Finally, validation-set checkpoint selection makes reported best scores selected rather than independent estimates.

Two collaborator-side artifacts remain. One numeric packet fills the matched DTEM, latency, memory, mass-bias, and recovery cells under a single protocol. One passive trace from the final checkpoint supplies routing visualizations with checks that tracing leaves logits, top- K indices, and mass conservation unchanged. Trained weights are outside the public-release scope, so the paper retains record-level provenance and does not claim public checkpoint replay. Until the numeric packet and trace arrive, the placeholders stay visibly unfilled and make no contribution to the reported results.

8 CONCLUSION

MergeNet combines continuous spatial mass transport with an exact-budget gather, so a learned routing policy feeds a physically shorter latent sequence in the same forward path. The original-grid prior consistently improves routing controls, including a degree-matched three-seed study, and accumulated carrier mass admits an exact FlashAttention-compatible key-bias construction. Current ImageNet results also expose the remaining gap: the model does not yet exceed dense DeiT-S/8 or post-training ToMe in accuracy, and its deployment frontier awaits common-protocol checkpoint measurements. This evidence supports the compression boundary and spatial-routing mechanism while defining a precise, testable path to a stronger system result.

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Table 5: Best EMA top-1 across resolution schedules. The gap is MergeNet minus dense in percentage points.

Pixels	Schedule	DeiT-S/8 (%)	MergeNet (%)	Gap (pp)
224	300 scratch	82.246	81.368	-0.878
384	300+30	82.976	82.582	-0.394
512	300+30+15	83.068	82.618	-0.450

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A TRAINING RECIPE AND EVIDENCE INVENTORY

Table 6 summarizes the resolved run arguments. The base width is 384 with six heads and MLP expansion four. Stochastic depth is configured at 0.1; MergeNet constructs separate local and latent ramps, unlike a single twelve-block dense ramp. The global and flattened routing controls use their dense/historical scoring paths, while radius-2/3 routing uses the sparse spatial backend; the routing support and its numerical implementation should therefore both be recorded. EMA validation replaces raw-model metrics before the summary row is written, so `eval_top1` is an EMA metric.

Setting	224-pixel scratch	384/512-pixel fine-tune
Epochs	150 or 300	30 at 384; 15 at 512
Effective global batch	1024	1024
Optimizer	AdamW	AdamW
Learning rate	5×10^{-4} or 7.5×10^{-4}	10^{-5}
Minimum learning rate	10^{-5}	10^{-6}
Weight decay	0.05	10^{-8}
Warmup epochs	5 (probe: 20)	1
Warmup learning rate	10^{-6}	10^{-6}
EMA decay	0.99996	0.99996
Mixup / CutMix	0.8 / 1.0	0 / 0
Label smoothing	0.1	0.1
Random erasing	0.25	0.25
RandAugment	<code>rand-m9-mstd0.5-inc1</code>	same
Gradient clip norm	1.0	1.0
Validation crop percentage	0.9	1.0
MergeNet auxiliary-logit weight	ramp to 0.05	0.05
Teacher / feature / routing distillation	disabled	disabled

Table 6: Main recipe. Launch records take precedence over configuration-file defaults, especially for microbatch size and gradient accumulation.

Classifier implementation. The main configuration uses patch size 8, width 384, six heads, six local blocks, and six latent blocks. At 224 pixels this gives 784 patch slots; the local attention band extends 16 sequence positions on either side of a patch query, while the CLS query attends globally. Spatial routing uses radius 3, a $384 \rightarrow 64$ metric projection, similarity temperature 1, and input-gradient scale 0.1 for the metric. At compression factor 2, the six soft donor budgets are (65, 65, 65, 65, 65, 67) and the hard gather retains 392 patches. There is no further merging in the latent encoder. Training partitions donors and receivers randomly for each image; evaluation alternates deterministic odd/even partitions across routing steps.

Selection and tracing. The threshold-based soft top- k uses scale 30 and normalizes row energies by their maximum-minus-minimum range across the current minibatch. Thus soft selection can depend on batch companions; the released operator has no guard for a zero range. The final hard-gather indices are detached and selected carriers are sorted by a detached, mass-weighted center of their flattened source indices. The default center-tracing mode does not retain a full source membership matrix. Spatial scoring uses sparse neighbor lists, although model initialization constructs a dense Boolean adjacency buffer.

During training, an auxiliary readout averages full-grid features using soft-selection weights, applies the shared classification head, and adds its weighted logits to the latent CLS logits. It is absent at evaluation and is not a separate auxiliary loss. The 150-epoch MergeNet runs start this contribution at epoch 10 and ramp it over 10 epochs; the 300-epoch run uses epoch 20 and a 20-epoch ramp. Fine-tunes enable weight 0.05 from the start. The recorded environment is Python 3.10.20, PyTorch 2.7.1+cu118, torchvision 0.22.1+cu118, timm 0.9.11, FlashAttention 2.7.3 and Triton 3.3.1 on H20-3e GPUs. Both four and eight ranks divide 50,000, avoiding validation-sampler padding for this split. Other world sizes require a separate check.

Three nominal launcher failures occurred after complete metric histories: the 150-epoch DeiT-S/16, the compound $\lambda = 4$ run, and dense 512-pixel fine-tuning. Complete summaries and supplied

checkpoint receipts distinguish them from early failures. Receipts record checks on another machine, not weights independently loaded in this audit. The λ curriculum and 512-pixel MergeNet fine-tune are now complete 150-epoch and 15-epoch records. The weight-decay probe was abandoned at 140/150. The DeiT-initialized and progressive-latent runs are complete 150-epoch records.

Run ID	Epochs	Status	Best top-1
s2_deit_p16_150e	150/150	Complete	77.196
s2_deit_p8_150e	150/150	Complete	79.992
s2_mn_flat_w8_150e	150/150	Complete	78.290
s2_mn_global_150e	150/150	Complete	77.218
s2_mn_l4_150e	150/150	Complete	78.366
s2_mn_r2_150e	150/150	Complete	78.942
s2_mn_r3_150e	150/150	Complete	78.888
s2_mn_r3_lr75_150e	150/150	Complete	79.316
s3_deit_p16_300e	300/300	Complete	79.692
s3_deit_p8_300e	300/300	Complete	82.246
s3_mn_r3_lr75_300e	300/300	Complete	81.368
s3_deit_p8_ft384_30e	30/30	Complete	82.976
s3_mn_r3_ft384_30e	30/30	Complete	82.582
s3_mn_r5_ft384_30e	30/30	Complete	82.574
s4_mn_lamcurr_150e	150/150	Complete	79.310
s4_mn_warm20_150e	150/150	Complete	79.238
s4_deit_ft512_15e	15/15	Complete	83.068
s4_mn_ft512_15e	15/15	Complete	82.618
s5_mn_dp07_150e	150/150	Complete	79.534
s5_mn_lr1e3_150e	150/150	Complete	79.350
s5_mn_wd03_150e	140/150	Partial	79.054
s6_deitB_p16_300e	300/300	Complete	80.442
s7_mn_deitinit_150e	150/150	Complete	80.574
s7_mn_proglatent_150e	150/150	Complete	79.392

Table 7: Status through the September 23 packet. The abandoned weight-decay metric is included for accounting only and is not an endpoint. The DeiT-B best score is a pre-overfitting peak (Appendix D). The companion data retain full precision and source summary lines.

A.1 INITIALIZATION AND LATENT-COMPRESSION DIAGNOSTICS

Table 8 uses the completed drop-path-0.07 run as the 150-epoch scratch control. Initializing MergeNet from the 300-epoch DeiT-S/8 checkpoint raises best EMA top-1 by 1.040 points. This result shows initialization sensitivity, not equal-total-compute superiority: the source representation already consumed 300 training epochs, the experiment has one seed, and its 80.574% endpoint remains below both the 300-epoch scratch MergeNet result (81.368%) and its dense source (82.246%). The original first-launch loading log was not delivered after a host migration; the record retains a contemporaneous 152-loaded/25-missing receipt and the early learning curve, so we treat this as a diagnostic rather than a primary causal result.

The progressive-latent variant adds 192 latent-stage merges and reaches 79.392%, 0.142 points below the same control. It changes the final configured budget from approximately 392 to 200 patches, so it is an accuracy-compression probe rather than a matched-budget test of gradual versus abrupt scheduling. With one seed, the result establishes no observed gain but does not establish a statistically resolved degradation. Training-log throughput is not used as deployment latency.

Table 8: Completed 150-epoch Stage-7 diagnostics, all with seed 42 and drop-path 0.07. Deltas use the scratch control in the first row.

Variant	Setting / change from control	Best (%)	Δ (pp)
Scratch control	drop-path 0.07	79.534	0.000
DeiT initialization	initialize from DeiT-S/8 300e	80.574	+1.040
Progressive latent	192 additional latent merges	79.392	-0.142

B DETAILED BUDGET AND TIMING COMPARISONS

Table 9 gives the values behind the main accuracy panels. The MergeNet checkpoint uses radius 3 at 224 pixels and radius 5.143 after 384-pixel fine-tuning. Final patch budgets match within a row, but layerwise attention and MLP inputs do not. ToMe and PiToMe shorten sequences within successive blocks, while MergeNet gathers once between stages. The companion CSV also contains the two resolution-transfer panels using 224-pixel checkpoints without fine-tuning.

Checkpoint / pixels	Patches	MergeNet	ToMe	PiToMe	Δ ToMe
224 trained / 224	523	81.380	82.038	81.876	+0.658
224 trained / 224	392	81.360	81.932	81.742	+0.572
224 trained / 224	314	81.074	81.814	81.560	+0.740
224 trained / 224	262	80.850	81.830	81.500	+0.980
224 trained / 224	196	80.228	81.752	81.206	+1.524
384 tuned / 384	1537	82.500	82.868	82.658	+0.368
384 tuned / 384	1152	82.562	82.866	82.534	+0.304
384 tuned / 384	923	82.230	82.800	82.452	+0.570
384 tuned / 384	770	82.002	82.792	82.334	+0.790
384 tuned / 384	576	81.300	82.742	82.296	+1.442

Table 9: Top-1 percentages at equal final patch counts; Δ is ToMe minus MergeNet in percentage points. These values cannot be paired with Table 10 to construct an accuracy–latency frontier.

ToMe patches	MN patches	ToMe ms	MN ms	ToMe GiB	MN GiB
1536	1538	147.30	178.17	2.793	3.619
1151	1152	129.81	155.25	2.764	3.564
922	923	122.44	143.02	2.746	3.721
769	770	118.17	136.59	2.738	3.719
575	576	106.45	129.38	2.723	3.557

Table 10: Separate 384-pixel synthetic latency experiment: batch 64, H20-3e, fp16 autocast, random initialization, MergeNet radius 5.143, and ToMe proportional attention disabled. Patch counts exclude the class token here. Memory is peak *reserved* GiB, unlike the historical 224-pixel record, which reports peak allocated memory. Token-counting hooks remain active during timing.

At the approximately half-size operating point, ToMe processes 1,151 patches and MergeNet 1,152, taking 129.810 and 155.255 ms respectively. These times do not prove a joint ordering for the accuracy sweep’s ToMe setting. A common harness should use exactly 1,152 patches with proportional attention enabled, verify the checkpoint and preprocessing, and remove shape hooks before timing. An earlier latency benchmark silently clamped several requested ToMe budgets to the same achieved count; those obsolete cells are excluded.

Separate 384-pixel radius-3 benchmarks remain distinct from the radius-5.143 sweep. Speed percentages at different batch sizes, radii or harness settings should not be averaged into an unexplained range. Peak reserved memory and peak allocated memory also measure different quantities.

C EVIDENCE AND REPRODUCTION BOUNDARIES

The artifact subset includes unmodified per-run summaries, resolved arguments with host paths removed, launch-protocol extracts, sanitized successful sweep records, audited benchmark tables, and source hashes. The CPU rebuild script recomputes best/final metrics, resolves sweep records by timestamp, checks numeric joins, and regenerates tables and figures. It never launches training. Source and claim ledgers retain their literature retrieval dates; the collaborator-packet audit was refreshed on September 21. Company policy prevents ImageNet weight transfer; independent ImageNet checkpoint re-evaluation is unavailable. The September 23 packet additionally completes the two Stage-7 diagnostics. The refreshed audit matches 3,354 of 3,560 summary rows to delivered EMA logs, leaving 206 rows without corresponding log records (Section 4). Those rows are not discarded, but retain their summary-only evidence status.

D DEiT-B/16 IS NOT A USABLE SCALE ANCHOR

A 300-epoch DeiT-B/16 run at 224 pixels completes under the same global-batch and seed protocol as the small-scale campaign, with drop-path 0.1 inherited from the DeiT-S configurations. Table 11 reports the peak and final EMA scores. Train loss continues to fall while evaluation loss rises after the peak at epoch 179. The final top-1, 78.566%, is below our DeiT-S/16 result of 79.692% at the same 196-token budget. Published DeiT-B 300-epoch training uses drop-path 0.2. We therefore do not append this run to the DeiT-S/8 table, and we do not launch a paired MergeNet-B comparison from these weights.

Epoch	Train loss	Eval loss	EMA top-1 (%)
150	2.968	0.889	80.060
179 (peak)	2.808	0.895	80.442
299 (final)	2.348	1.095	78.566

Table 11: DeiT-B/16, 300 epochs, drop-path 0.1. Peak and final scores must be reported together; the peak is not a converged result.

E A BOUNDED LAYOUT OPTIMIZATION EXPERIMENT

We additionally test a single LocalAttention module on an A100-SXM4-80GB. The original module constructs query/key/value tensors in batch-head-token-channel order, then transposes and copies them for the FlashAttention wrapper. An isolated variant constructs batch-token-head-channel tensors directly and adjusts the global class-query attention views and final reshape accordingly. It does not alter the released model, load a checkpoint, or apply an optimizer update.

The test uses PyTorch 2.6.0+cu124 and FlashAttention 2.7.4.post1, width 384, six heads, window radius 16, zero attention dropout, float32 parameters and inputs, and fp16 autocast. Four batch-2 parity cases cover sequence lengths 785 and 2,305, global class attention on/off, and query/key normalization. Outputs and input gradients are bitwise equal in these cases. Parameter gradients are also equal except for the normalization case, whose maximum relative L_2 error is 5.77×10^{-7} . This coverage does not establish all-shape or dropout/RNG-preserving equivalence for a complete model.

Batch	Tokens	Mode	Original ms	Layout ms	Reduction range
64	785	Forward	1.5600	1.0448	13.50–34.03%
64	785	Forward + backward	5.1590	4.3999	14.68–25.64%
16	2305	Forward	1.0318	0.9369	6.88–36.68%
16	2305	Forward + backward	5.3740	4.9176	5.45–11.83%

Table 12: Single-module A100 measurements. Times are medians of three round medians, each with five warmups and twenty CUDA-event samples. The final column is the range of paired-round time reductions, not a confidence interval. Tokens include the class token. Backward includes a squared-output loss and gradient clearing, with no optimizer.

Measurement order alternates by round, and the harness requires an initially empty device, establishes its CUDA process identity, then rejects a changed process set at subsequent checks. All iteration samples and snapshots are retained. The observed reductions support this local engineering direction, but the round variation is appreciable. Both module instances are resident, so their recorded memory peaks are not independent per-model memory measurements. These results do not predict the full model’s speedup, validate an HBM-traffic estimate, transfer H20 timings to A100, or resolve the separate ToMe attention-policy mismatch.

F RESERVED TRAINED-CHECKPOINT ROUTING VISUALIZATION

The final ImageNet checkpoint is unavailable locally. The collaborator will export four to six validation examples with the actual donor/receiver partition, nonzero transport edges, token-mass maps,

final carriers, and at least one failure case. The trace is passive: it records maximum logit difference with tracing enabled, equality of top- K indices, and mass conservation error after every step. Without a source-membership trace, the figure will show per-step transport rather than an invented final segmentation.

P06: trained images, routing edges, carriers, and a failure case
Reserved space; no measurements or images supplied.

Figure 2: Reserved trained-checkpoint routing visualization. The blank is not a model observation and will be replaced only by a verified trace.

G HISTORICAL CIFAR-100 SPATIAL-ROUTING ABLATION

We retain a historical CIFAR-100 experiment as a limited diagnostic of the routing geometry. The August campaign contains 60 new runs: two MergeNet configurations, five input sizes, and six routing settings (Global and Euclidean radii $R \in \{1, 2, 3, 4, 8\}$), all using seed 42. The configurations have local/latent depths of 6/6 with final compression ratio $\lambda = 2$, or 4/8 with $\lambda = 4$; both use width 384, six heads, and 8×8 patches. Each model was trained from scratch for 200 epochs without distillation. Training used AdamW, learning rate 10^{-3} , weight decay 0.05, a cosine schedule with 20 warmup epochs, effective batch size 200, and EMA decay 0.9998. The primary endpoint is the final epoch-199 EMA top-1 accuracy on the 10,000-image CIFAR-100 test set. The frozen local campaign protocol designated $R = 3$ as the primary setting before the recorded training launches; this is an internal protocol record, not an external preregistration. We keep that setting even though $R = 2$ has a slightly higher mean.

The original images are 32×32 pixels and are resized to the listed input sizes. Thus, this sweep changes the patch grid over upsampled images and does not test native high-resolution image content. Evaluation used FP16 autocast and deterministic fast grouping, with batches of 200, 200, 200, 100, and 50 at sizes 160, 192, 224, 256, and 320, respectively. Global and the finite-radius settings were trained in the same frozen implementation and share the non-routing configuration. The additional Flat reference uses the historical flattened-window implementation with width parameter 8 and comes from ten earlier August runs. Its recorded non-routing training arguments and dependency versions match, but its runtime code snapshot differs; three Flat runs resumed within their original jobs. We therefore use Flat as a historical descriptive reference.

Configuration	Input	Global	Flat [†]	$R = 3$	ΔG	ΔF
6/6, $\lambda = 2$	160	67.80	68.79	69.42	+1.62	+0.63
	192	69.42	69.93	70.85	+1.43	+0.92
	224	69.50	70.53	70.53	+1.03	+0.00
	256	69.50	70.12	71.03	+1.53	+0.91
	320	69.85	70.81	71.42	+1.57	+0.61
4/8, $\lambda = 4$	160	65.29	66.69	67.92	+2.63	+1.23
	192	69.87	69.75	71.27	+1.40	+1.52
	224	68.21	69.87	70.52	+2.31	+0.65
	256	67.35	68.30	69.40	+2.05	+1.10
	320	66.05	67.08	68.83	+2.78	+1.75
Mean		68.284	69.187	70.119	+1.835	+0.932

Table 13: Historical CIFAR-100 final-epoch EMA accuracy (%) and $R = 3$ deltas (percentage points) against Global (ΔG) and Flat (ΔF). Each row uses seed 42; the mean gives equal weight to the ten configuration–input pairs. [†]Flat is a historical reference with a different code snapshot. Input denotes the resized side length of native 32×32 images. These are observations under the stated evaluation batches and historical dense-score implementation.

At the fixed $R = 3$ setting, accuracy exceeds Global in all ten pairs, with a mean improvement of 1.835 percentage points. Relative to historical Flat, nine pairs improve and one ties, with a mean difference of 0.932 points. Across the complete sweep, the mean accuracies for Global and radii 1, 2, 3, 4, and 8 are 68.284, 68.774, 70.225, 70.119, 70.047, and 68.414%, respectively. Radii 2, 3, and 4 each exceed their matched Global control in all ten pairs. These correlated, single-seed observations provide evidence that restricting routing neighborhoods can help in this particular setting; they do not establish statistical significance, native-resolution transfer, or superiority over external token-reduction methods.

The audit reconciled all 60 new endpoints and the ten Flat endpoints with raw 200-epoch summaries, completion receipts, and frozen manifests. Archived full-test evaluations of the 60 new epoch-199 EMA checkpoints show identical generic- and fast-grouping logits; their accuracies differ from the training summaries in 31 cases by at most 0.02 points. The table preserves the registered training-summary endpoint. The initial historical audit checked receipt hashes and checkpoint presence, without freshly evaluating or rehashing all checkpoints. The historical spatial implementation constructs a dense donor–receiver score matrix before masking, so these results do not validate the later sparse backend or its efficiency. Section 5.2 reports a later 12-run campaign with three seeds and

a degree-matched control; those runs are independent trainings and are not substitutes for the table above.

H INDEPENDENT FOLLOW-UP WITH AVAILABLE LOCAL ARTIFACTS

Company policy prevents transfer of the ImageNet checkpoints. We therefore separate two feasible follow-ups: architecture timing using random weights, and prediction tests using existing, locally available CIFAR-100 weights. Neither experiment independently reproduces the company ImageNet scores.

H.1 A COMMON A100 INFERENCE HARNESS

We compare dense DeiT-S/8, ToMe, and the canonical sparse-backend MergeNet on an A100-SXM4-80GB using PyTorch 2.6.0+cu124, timm 0.9.11 and FlashAttention 2.7.4.post1. All cells use batch 64, float32 parameters and inputs, and fp16 autocast. The compressed models retain exactly 392 patches at 224 pixels and 1152 at 384 pixels, excluding CLS. A diagnostic forward records block outputs; all hooks are removed before timing. ToMe source tracing is disabled. Proportional attention is explicitly controlled, with its enabled setting matching the recorded accuracy protocol.

Each cell has ten warm-up iterations and thirty CUDA-event samples in each of three rounds, reversing model order in the middle round. A selected-GPU process guard requires an empty device before setup and only our process during the experiment. Models are allocated and released sequentially. Table 14 reports the median of three round medians. It measures model execution, excluding data loading, backward and updates.

Model	224 px (ms)	384 px (ms)
Dense DeiT-S/8	30.09	109.43
ToMe, proportional attention on	34.42	133.36
ToMe, proportional attention off	29.71	96.05
MergeNet $R = 3$	49.16	144.97
MergeNet $R = 3$, BNHD	46.56	138.11
MergeNet $R = 5.143$	–	218.25
MergeNet $R = 5.143$, BNHD	–	211.40

Table 14: Independent synthetic A100 inference, batch 64. All weights are randomly initialized; these are not trained-checkpoint frontier points. BNHD changes the local-attention tensor layout. The 384-pixel radius-5.143 model has the same final patch budget as radius 3.

On this local hardware/software stack, the original MergeNet is slower than dense DeiT at both resolutions. The BNHD change reduces complete-model radius-3 time by about 5%, considerably less than the isolated-module improvement. The proportional-attention setting materially affects ToMe latency. Thus neither the H20 crossover nor a ToMe timing measured with proportional attention disabled should be treated as accelerator-independent evidence. The A100 comparison changes the hardware and software stack relative to the H20 experiment; it does not isolate a hardware-only cause. No historical accuracy point is attached to these random-weight timings.

H.2 LAYOUT VALIDATION UNDER A COMPLETE TRAINING GRAPH

The layout comparison uses identical weights and images, restored CPU and CUDA random states, fresh per-sample routing, and zero dropout at both 224 and 384 pixels. With the dense routing reference backend, outputs, input gradients and all parameter gradients are bitwise equal. With the sparse backend and an unscaled squared-logit loss, the strict gradient comparison fails; an unchanged-model repeat also fails, so these runs do not isolate an error introduced by BNHD. Their input-gradient relative errors are approximately 4.8% and 16.3%, respectively.

We retain these failed tests and add a fixed loss scale of 1024, unscaling all gradients before comparison. The sparse-backend tests then pass the predefined relative- L_2 limits of 0.002 for outputs and 0.003 for gradients. Maximum parameter-gradient relative errors are below 0.0016; unchanged-model repeat controls are below 0.0018. This diagnoses sensitivity of the unscaled fp16 test and

supports bounded layout parity. It does not establish long-run training equivalence, eliminate sparse-kernel nondeterminism, or validate arbitrary dropout and precision settings.